

5.1 Vector Fields

Up to this point in MTH 202 we have studied functions of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$ and $f: \mathbb{R}^n \rightarrow \mathbb{R}$. In this final chapter of MTH 202 we turn our focus to the most general situation. Namely we will study functions of the form $\mathbf{F}: \mathbb{R}^m \rightarrow \mathbb{R}^n$.

Definition:

1. Let D be a set in $\mathbb{R}^2$. A vector field on $\mathbb{R}^2$ is a function $\mathbf{F}$ that assigns to each point (x, y) in D a two-dimensional vector $\mathbf{F}(x, y)$.
2. Let E be a set in $\mathbb{R}^3$. A vector field on $\mathbb{R}^3$ is a function $\mathbf{F}$ that assigns to each point (x, y, z) in E a three-dimensional vector $\mathbf{F}(x, y, z)$.

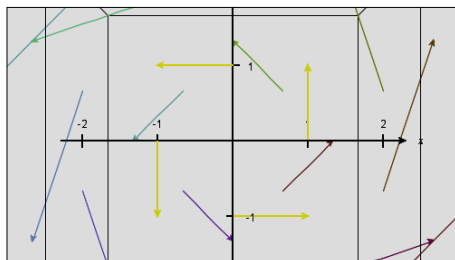
Example 1: Use a picture to illustrate the vector field $\mathbf{F}(x, y) = \langle -y, x \rangle$.

In order to sketch a picture of this vector field, we plug points (x, y) into $\mathbf{F}$. Then we draw the vector that $\mathbf{F}$ spits out on the xy -plane with initial point (x, y) .

So, we note that:

(x, y)	$\mathbf{F}(x, y)$
$(0, 0)$	$\langle 0, 0 \rangle$
$(1, 0)$	$\langle 0, 1 \rangle$
$(0, 1)$	$\langle -1, 0 \rangle$
$(-1, 0)$	$\langle 0, -1 \rangle$
$(0, -1)$	$\langle 1, 0 \rangle$
$(1, 1)$	$\langle -1, 1 \rangle$
$(-1, -1)$	$\langle 1, -1 \rangle$

Now, a picture illustrating $\mathbf{F}$ is:



Remark: Vector fields have a wide variety of applications. For example, a vector field on $\mathbb{R}^2$ can be used to model the amount of force exerted by an electric charge on some charge located at the point (x, y) on a two dimensional plate.

A vector field on $\mathbb{R}^3$ can be used to model the force of the wind at some point (x, y, z) in space.

A vector field on $\mathbb{R}^3$ can be used to model the current of the ocean at some point (x, y, z) in the ocean.

A vector field on $\mathbb{R}^3$ can be used to model the force due to gravity at some point (x, y, z) in space.

A vector field in $\mathbb{R}^3$ can be used to model the velocity of a moving fluid at some point (x, y, z) in space.

Example 2: The gravitational force from an object of mass M centered at the origin acting on a small object of mass m at $\mathbf{x} = \langle x, y, z \rangle$ is $\mathbf{F}(x, y, z) = -\frac{mMG}{\|\mathbf{x}\|^3} \mathbf{x}$ where G is the universal gravitational constant.

Suppose a large three dimensional object has center of mass at the origin and has mass equal to 100,000 kg. Suppose a small object of mass 5 kg is located at the terminal point of the position vector $\mathbf{x} = \langle 10, 5, 10 \rangle$ (you may assume distance is in meters). What is the gravitational force acting on the small object (due to the large object)?

It is $\mathbf{F}(10, 5, 10) = -G \left(\frac{500000}{\|\langle 10, 5, 10 \rangle\|^3} \right) \langle 10, 5, 10 \rangle = \frac{-4000G}{27} \langle 10, 5, 10 \rangle = \left\langle \frac{-40000G}{27}, \frac{-20000G}{27}, \frac{-40000G}{27} \right\rangle$. The unit on this force would be in Newtons.

Note: Since $G \approx 6.67 \cdot 10^{-11}$, we see that the force calculated in the above example is quite small.

Definition: If f is a scalar function of two or three variables, $\vec{\nabla} f$ is called the gradient vector field of f .

Definition: A vector field $\mathbf{F}$ is called a conservative vector field if it is the gradient of some scalar function f . Mathematically, this means there must exist a scalar function f so that $\mathbf{F} = \vec{\nabla} f$. In this situation f is called a potential function for $\mathbf{F}$.

Example 3: Prove that $\mathbf{F}(x, y) = \langle x, y \rangle$ is a conservative vector field.

Consider the scalar function $f(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$. We note that $\vec{\nabla} f(x, y) = \langle x, y \rangle = \mathbf{F}(x, y)$. Thus, $\mathbf{F}(x, y) = \langle x, y \rangle$ is a conservative vector field by definition. ■

Question: Is $f(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$ the only scalar function that would have worked in the above proof?

Note: For now, the only technique we have for proving a vector field is conservative is to guess and check until we successfully find a potential function for the vector field. We will develop more sophisticated techniques as in section 5.3.

Note: Right now, we do not have ANY technique for proving that a vector field is not conservative. We will develop techniques for proving a vector field is not conservative in section 5.3.

Example 4: Consider $f(x, y, z) = xye^z$.

a) Find the gradient vector field of f .

It is $\vec{\nabla} f(x, y, z) = \mathbf{F}(x, y, z) = \langle ye^z, xe^z, xye^z \rangle$.

b) Is the vector field you found in (a) conservative? Explain.

Yes, it is conservative because it is the gradient of the scalar function $f(x, y, z) = xye^z$.

Note: Later in chapter 5 we will discover that conservative vector fields have some very nice properties.